

Solution

LIGHT -05

Class 10 - Science

1.

(c) $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Explanation: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

2. (a) between focus and centre of curvature

Explanation: between focus and centre of curvature

3.

(b) on the same walls as the window

Explanation: Since reflection of light received from the distant object can be received only then.

4.

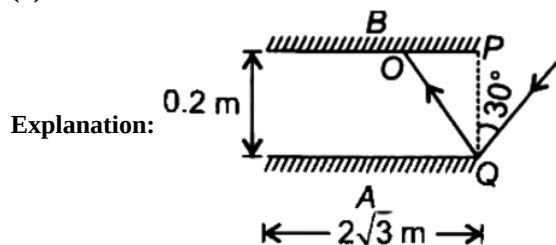
(d) 25 cm

Explanation: Radius of curvature $R = 2f$

$$R = 2 \times 12.5 = 25 \text{ cm}$$

5.

(b) 30



According to laws of reflection, $\angle i = \angle r$

So, $\angle PQO = 30^\circ$

Given: $PQ = 0.2 \text{ m}$

$$\tan 30^\circ = \frac{OP}{PQ}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{OP}{0.2} \Rightarrow OP = \frac{0.2}{\sqrt{3}} \text{ m}$$

OP is distance covered by the ray in one reflection. Number of times the ray undergoes reflections, before

$$\text{it emerges out} = \frac{2\sqrt{3}}{OP} = \frac{2\sqrt{3}}{\frac{0.2}{\sqrt{3}}} = \frac{6}{0.2} = 30$$

6.

(b) When object is placed between the focus and centre of curvature

Explanation: When an object is placed between F and C an enlarged image is formed beyond C.

7.

(c) Concave in nature

Explanation: Concave in nature

8.

(d) 1,1

Explanation: We know, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

Here, $f = -10 \text{ cm}$, $u = -15 \text{ cm}$

$$\therefore \frac{1}{-10} = \frac{1}{-15} + \frac{1}{v}$$

$$\text{or, } \frac{1}{-10} = \frac{1}{-15} + \frac{1}{v} = -\frac{1}{30}$$

or, $v = -30 \text{ cm}$

Hence, two images will be formed at a distance of 30 cm from mirror.

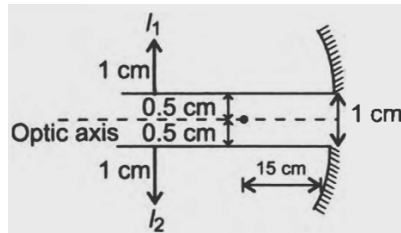
$$\text{We know, } m = -\frac{v}{u} = -\frac{(-30)}{(-15)} = -2$$

Distance between broken mirrors is 1 cm

$$h_{i1} = mh_{o1} = (-2) \times (-0.5) = 1 \text{ cm}$$

Image (I_1) due to upper part of the mirror will be formed at 1 cm above its optic axis.

Similarly, image (I_2) due to lower part of the mirror will be formed at 1 cm below its optic axis



9.

(c) is less than one

Explanation: The convex mirror is used in the rearview mirror. The convex mirror always gives a smaller image. Hence, magnification produced by the rearview mirror is always less than 1.

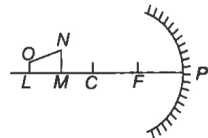
10.

(c) Both $|m_{LO}| < 1$ and $|m_{MN}| < 1$ and $|m_{LO}| < |m_{MN}|$

Explanation: Object is placed beyond C. Hence, the image will be real and it will lie between C and F. Further u and v are negative, hence the mirror formula will become

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{u-f}{uf} \quad \text{or } v = \frac{f}{1-(f/u)}$$



Now $u_{LO} > u_{MN} \therefore v_{LO} < v_{MN}$

$|m_{LO}| < |m_{MN}|$ (As $m = -\frac{v}{u}$)

Note that, $u_{LO} < u_{LO}$ and $v_{MN} < u_{MN}$

So, $|m_{LO}| < 1$ and $|m_{MN}| < 1$.

11. (a) More than 16 metres

Explanation: The image of R is 8 m behind the mirror (say R') perpendicularly. However, with respect to camera at M, it is the diagonal distance (MR'), which is more than 16 m.

12.

(d) -15 cm

Explanation: Given the magnification +2.0 of a concave mirror of radius of curvature 60.0 cm.

$$\text{Focal length} = \frac{60}{2} = -30$$

(-ve is taken because the focus of the mirror is behind the pole.)

$$\text{Magnification, } m = \frac{-v}{u}$$

Putting the value of magnification in the above formula we get,

$$\frac{-v}{u} = 2$$

$$v = -2u$$

v is image distance, u is object distance and f is focal length.

Since,

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

Putting the values

$$\frac{1}{u} + \frac{1}{-2u} = \frac{1}{-30}$$

$$\frac{1}{2u} = \frac{1}{-30}$$

$$u = -\frac{30}{2} \text{ cm}$$

$$u = -15 \text{ cm}$$

Hence, the object distance is -15 cm.

13. (a) a convex mirror

Explanation: a convex mirror

14. (a) Between F and 2F

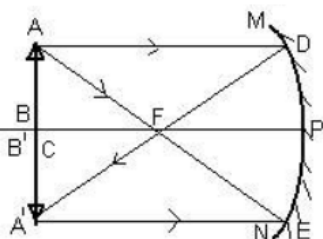
Explanation: When an object is placed at a distance more than twice the focal length (F) from a concave mirror, the image will be formed at a distance between F and 2F.

15. (a) between pole and focus

Explanation: between pole and focus

16. (a) 30 cm in front of the mirror

Explanation:



The ray diagram illustrates that, in the case of a concave mirror, the image size becomes equal to the object size when the object distance equals the radius of curvature, i.e, twice the focal length.

So, the object has to be placed at a distance of $(15 \times 2) = 30$ cm in front of the concave mirror.

17. (d) 2.5 cm
Explanation: Since the distance of the object from the pole of the mirror is equal to the radius of curvature of the mirror, the object is kept at the centre of curvature of the mirror. When an object is kept at the centre of curvature of a concave mirror, the image is also formed at the centre of curvature.

According to the mirror formula,

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

Given, $u = -2f$ and $f = -f$

$$\therefore \frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{(-f)} - \frac{1}{(-2f)}$$

$$\therefore \frac{1}{v} = \frac{1}{2f} - \frac{1}{f} \text{ or } v = -2f$$

$$m = \frac{\text{Height of the image (h')}}{\text{Height of the object (h)}} = \frac{\text{Image distance (v)}}{\text{Object distance (u)}} = 1$$

The size of the image is the same as that of the object i.e. 2.5 cm

18. (b) 3 m

Explanation: 3 m

19. (d) when the object placed between F and C
Explanation: when the object placed between F and C

20. (b) at the centre of curvature
Explanation: at the centre of curvature

21. (d) 16 cm
Explanation: Radius of curvature, $R = 32$ cm

Focal length, $f = ?$

We know that,

$$f = \frac{R}{2}$$

$$f = \frac{32}{2} = 16\text{cm}$$

22. (d) 6 cm
Explanation: $u = -15$ cm, $v = -10$ cm
 $f = ?$
 We know that,

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{(-10)} + \frac{1}{(-15)} = \frac{1}{f}$$

$$\Rightarrow \frac{1}{f} = -\frac{1}{10} - \frac{1}{15} = \frac{-3-2}{30} = -\frac{5}{30} = -\frac{1}{6}$$

$\therefore f = -6 \text{ cm}$

$\therefore$ The focal length of the concave mirror is 6 cm.

- 23.
- (d)** $u = -1.05 \text{ cm}$ and $m = 19.05 \text{ cm}$
Explanation: $u = -1.05 \text{ cm}$ and $m = 19.05 \text{ cm}$
- 24.
- (b)** very near to the focus of the reflector.
Explanation: In torches, search lights and headlights of vehicles concave mirrors are used. When a light source is placed at the focus of the mirror, after reflection light rays become parallel to the principal axis. Due to this, the bulb is placed very near to the focus of the reflector.
- 25.
- (b)** 2.4
Explanation: 2.4
- 26.
- (a)** convex mirror
Explanation: Both plane and convex mirror produce a virtual image. However, the image of a convex mirror is diminished always irrespective of the position of the object.
- 27.
- (d)** 50 cm
Explanation: Focal length (f) = 25 cm
 Radius of curvature (R) = ?
 We know that
 $f = \frac{R}{2}$
 $25 = \frac{R}{2}$
 $R = 25 \times 2$
 $R = 50 \text{ cm}$
- 28.
- (b)** at a distance greater than radius of curvature
Explanation: at a distance greater than radius of curvature
- 29.
- (c)** Concave mirror as well as convex lens
Explanation: Concave mirror as well as convex lens can make a parallel beam of light when light from a point source is incident on it.
- 30.
- (a)** 5 m
Explanation: New distance between the man and the mirror = $y + 5$
 $\therefore$ The distance between the man and his image
 $(y + 5) \times 2 = 20$
 $2y + 10 = 20$
 $2y = 10$
 $y = 5 \text{ m}$
- 31.
- (a)** virtual and erect
Explanation: When an object is kept within the focus of a concave mirror, an enlarged image is formed behind the mirror. This image is virtual and erect.
- 32.
- (c)** 0.5 m
Explanation: focal length = ?
 Radius of curvature, $R = 1 \text{ m}$ (+ for convex mirror)

As $f = \frac{R}{2}$

$\therefore f = \frac{1}{2} = 0.5 \text{ m}$

33.

(c) Concave mirror, convex mirror, concave lens and convex lens

Explanation: In case of all of the above, when an object is at infinity, image is highly diminished and point sized.

34.

(d) Black absorbs all the colours

Explanation: Black absorbs all the colours

35.

(c) Both concave

Explanation:

The mirror and the lens, both are concave. The distance of the principal focus measured from the pole is called the focal length. According to the New Cartesian Sign Convention, the focal length of a concave mirror as well as a concave lens is considered negative.

36.

(d) $R = 2f$

Explanation: $R = 2f$

37. (a) at infinity

Explanation: at infinity

38.

(d) sun

Explanation: Larger the distance, better and sharper focus can be achieved.

39.

(d) 15 cm

Explanation: We know, $m = \frac{h_i}{h_o} = \frac{-v}{u}$

Here, $m = 3$, or, $-m = \frac{v}{u}$

So, $m = \frac{-v}{u}$ or, $-m = \frac{v}{u}$

or $-m - 1 = \frac{v}{u} - 1$ or $-m - 1 = \frac{v-u}{u}$

or $-3 - 1 = \frac{40}{u}$ or $-4 = \frac{40}{u}$

or $u = -10 \text{ cm}$

$\therefore v - u = 40 \text{ cm}$

or $v + 10 = 40 \text{ cm}$

We know, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

So $f = \frac{uv}{u+v}$

or $f = \frac{-300}{20} = -15 \text{ cm}$

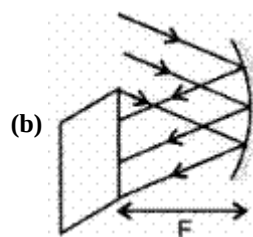
Thus, magnitude of focal length of the mirror is 15 cm.

40.

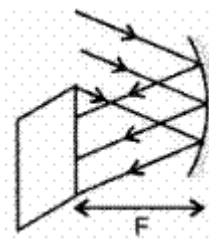
(b) at the focus of reflector

Explanation: at the focus of reflector

41.



Explanation:



42.

(b) 1.5cm

Explanation: 1.5cm

43.

(a) 120°

Explanation: Since, angle of incidence = 90° - angle between plane mirror and incident ray = $90^\circ - 30^\circ = 60^\circ$

and according to first law of reflection, angle of incidence = angle of reflection = 60°

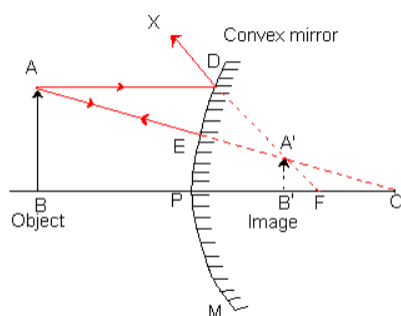
Total angle between incident ray and reflected ray = $60^\circ + 60^\circ = 120^\circ$

44.

(d) a convex mirror

Explanation:

The field of a convex mirror is more than any type of mirror. Hence the full-length size of the building can be seen by using a convex mirror.



45.

(b) A convex lens of focal length 5 cm

Explanation:

A convex lens (converging lens) of focal length 5 cm should be used to read small letters in a dictionary. A convex lens of short focal length (5 cm) bends the light rays through large angles, by focussing them closer to the optical centre.

46.

(d) 0 degree

Explanation: 0 degree

47.

(c) 4.5 cm

Explanation: Distance between the man and the mirror = 2 cm

Distance between man and chart = 50 cm = 0.5 m

Distance between chart and mirror = 0.5 m + 2 m = 2.5 m

Distance between the mirror and the image of the chart = 2.5 m

Distance between man and the image of chart = Distance between man and the mirror + Distance between the mirror and the image of the chart = 2 m + 2.5 m = 4.5 m

48.

(b) Very near to the focus

Explanation:

In torches, search lights and head lights of vehicles, the bulb is placed very near to the focus. Since the reflector of torches, search lights and head lights are concave in shape; the reflected light forms a parallel beam and travels a large distance.

49. (a) - 20 cm

Explanation: Here, size of object = O = +10.0 mm = +1.0 cm (as, 1 cm = 10 mm)

Size of Image size = I = 5.0 mm = 0.5cm

Image distance, v = -30 cm (as image is real)

Let. object distance = u

Focal length, f = ?

$$\text{Magnification } m = \frac{1(\text{Size of image})}{O(\text{Size of object})}$$

Magnification is given by $m = \frac{-v}{u}$

$$\frac{I}{O} = \frac{-v}{u}$$

$$\frac{0.5}{1} = \frac{-30}{u}$$

$$U = -60 \text{ cm}$$

Focal length is given by $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{1}{f} = \frac{1}{-30} + \frac{1}{-60} = \frac{-2-1}{60}$$

$$= \frac{-3}{60}$$

$$F = -20 \text{ cm}$$

50. (a) 0°

Explanation: The angle of reflection for the ray passing through the centre of curvature will be 0° . A ray passing through the centre of curvature of a concave mirror is reflected back along the same path.

51.

(c) Real and Inverted

Explanation: Real and Inverted

52. (a) Decreased after each reflection

Explanation: After each reflection, some part of light get absorbed by the mirror while most of the part is reflected. Thus, multiple reflections decreases the intensity of the image so only a few images are visible.

53.

(c) focus and centre of curvature

Explanation: For objects between f and 2f or centre of curvature, the image formed will be enlarged and beyond C.

54.

(b) ∞

Explanation: Image of an object at 2f will be at 2f, while it will be at ∞ , if object is at f.

55.

(b) more than 200 mm

Explanation: Since the image formed is upright, the object lies within the focus of the concave mirror.

So, $f > 100 \text{ mm}$

We know $R = 2f$

So, $R > 200 \text{ mm}$

i.e. Radius of curvature is more than 200 mm.

56.

(b) 15cm

Explanation: 15cm

57.

(c) in case (i) but not in case (ii)

Explanation: The mirror reflects while the lens refracts the light.

58.

(d) 7.5 m

Explanation: The reason being, the image formed by a plane mirror is the same distance behind the mirror as it is between the object and the mirror. A distance of 5 m between man and his image means that the distance between him and the mirror = $\frac{5}{2} = 2.5 \text{ m}$

Thus, the distance he should walk = $10 - 2.5 = 7.5 \text{ m}$

59. (a) towards the screen

Explanation: The blurred image is seen only when the separation between the screen and mirror is less than the focal length.

60.

(c) 4 cm

Explanation: Initially, the object is placed 20 cm in front of the mirror. So the image will form 20 cm behind the mirror.

If the mirror is moved 2 cm towards the object,

distance between object and mirror = $20\text{ cm} - 2\text{ cm} = 18\text{ cm}$

distance between mirror and image = 18 cm, which is 2 cm less than the initial case.

Since the mirror has also moved 2 cm away from the position of the original image, so the total distance between the positions of the original and final images is $2\text{ cm} + 2\text{ cm} = 4\text{ cm}$

61.

(c) retrace its path along the principal axis

Explanation: The angle of incidence for a ray along the principal axis is zero and so will be the reflection angle. So the ray will retrace its path.